Techfest 2025-26

InsightX: Leadership Analytics Challenge

1. Context

In the rapidly evolving digital payments ecosystem, organizations generate millions of transactions daily. While this data holds immense strategic value, extracting meaningful insights remains challenging for non-technical stakeholders. Business leaders need quick answers to questions like "Which merchant categories show the highest failure rates during peak hours?" or "How do transaction patterns differ between age groups on weekends?" without requiring expertise in data analytics or SQL.

A conversational insight system that understands natural language queries and responds with accurate, explainable answers can democratize data access and accelerate decision-making across product, operations, marketing, and risk management teams.

2. Objective

Design and develop a conversational AI system that allows users to query digital payment transaction data using natural language and receive intelligent, contextual responses. The system should:

- Understand diverse business questions related to transaction patterns, user behavior, operational metrics, and risk indicators
- Generate accurate insights backed by data analysis
- Provide clear explanations of findings, including relevant statistics and trends
- Support follow-up questions and contextual conversations
- Present insights in user-friendly formats (text summaries, statistics, recommendations)

3. Dataset Description

Column Name	Description
`transaction_id`	Unique identifier for each transaction
`timestamp`	Date and time of transaction
`transaction_type`	Type of transaction: P2P (Person-to-Person), P2M (Person-to-Merchant), Bill Payment, or Recharge
`merchant_category`	Category for P2M transactions: Food, Grocery, Fuel, Entertainment, Shopping, Healthcare, Education, Transport, Utilities, Other; **NULL for P2P transactions**
`amount_inr`	Transaction amount in Indian Rupees
`transaction_status`	SUCCESS or FAILED
`sender_age_group`	Age group of sender: 18-25, 26-35, 36-45, 46-55, 56+
`receiver_age_group`	Age group of receiver; **only applicable for P2P transactions, NULL otherwise**
`sender_state`	Indian state of the sender
`sender_bank`	Sender's bank: SBI, HDFC, ICICI, Axis, PNB, Kotak, IndusInd, Yes Bank
`receiver_bank`	Receiver's bank (same list as sender_bank)
`device_type`	Device used: Android, iOS, Web
`network_type`	Network connection: 4G, 5G, WiFi, 3G

`fraud_flag`	Binary indicator: 0 = not flagged, 1 = flagged for review; **NOTE: This represents transactions flagged for review, NOT confirmed fraud cases**
`hour_of_day`	Hour of transaction (0-23), derived from timestamp
`day_of_week`	Day of transaction (Monday-Sunday), derived from timestamp
`is_weekend`	Binary indicator: 0 = weekday, 1 = weekend

Participants will work with a **synthetic dataset of 250,000 digital payment transactions** representing realistic payment scenarios across India. This dataset has been created specifically for this hackathon and does not contain any real user data.

Dataset Schema

Important Notes

- **Fraud Flag Interpretation**: The `fraud_flag` column indicates transactions flagged by automated systems for manual review. A value of 1 does not confirm fraud but suggests anomalous patterns requiring investigation.
- **Derived Fields**: `hour_of_day`, `day_of_week`, and `is_weekend` are pre-computed from the `timestamp` field for convenience.
- **NULL Values**: `merchant_category` is NULL for P2P transactions; `receiver_age_group` is NULL for non-P2P transactions.

4. Expected Tasks

Participants are expected to:

1. **Data Understanding**: Analyze the provided dataset to understand distributions, patterns, and relationships between variables
2. **Query Understanding Module**: Build a natural language processing component that interprets user queries and identifies:
    - Intent (e.g., trend analysis, comparison, aggregation, anomaly detection)

    - Entities (e.g., specific transaction types, time periods, age groups, states)
    - Required computations or filters
3. **Insight Generation Engine**: Develop logic to:
    - Query and analyze the dataset based on user questions
    - Perform statistical analysis, aggregations, and pattern recognition
    - Identify trends, anomalies, and correlations
    - Generate meaningful responses with supporting data
4. **Conversational Interface**: Create a chat-based interface that:
    - Accepts natural language queries
    - Maintains conversation context for follow-up questions
    - Presents insights in clear, concise language
    - Handles ambiguous or incomplete queries gracefully
5. **Explainability**: Ensure all insights are explainable with:
    - Clear reasoning behind conclusions
    - Relevant statistics and metrics
    - Appropriate visualizations or data summaries where helpful

5. Expected Outputs

Core Deliverables

1. **Functional Conversational System**: A working prototype that can answer diverse business questions about the transaction data
2. **Query Coverage**: The system should handle questions across multiple categories:
    - **Descriptive**: "What is the average transaction amount for bill payments?"
    - **Comparative**: "How do failure rates compare between Android and iOS users?"
    - **Temporal**: "What are the peak transaction hours for food delivery?"
    - **Segmentation**: "Which age group uses P2P transfers most frequently?"
    - **Correlation**: "Is there a relationship between network type and transaction success?"
    - **Risk Analysis**: "What percentage of high-value transactions are flagged for review?"
3. **Response Quality**: Answers should include:

- Direct response to the question
- Supporting statistics and metrics
- Contextual insights or patterns
- Recommendations where applicable

Sample Interaction Example

User: "Which transaction type has the highest failure rate?"

System: "Bill Payments have the highest failure rate at 8.2%, followed by Recharges at 6.7%. P2M transactions have a 4.3% failure rate, while P2P transfers show the lowest failure rate at 2.1%. The higher failure rates in bill payments may be attributed to third-party biller system integrations and validation delays."

6. Evaluation Criteria

Solutions will be evaluated based on:

1. Insight Accuracy (30%)

- Correctness of data analysis and computations
- Accuracy of patterns and trends identified
- Validity of conclusions drawn from data

2. Query Understanding (25%)

- Range of query types handled
- Ability to parse complex or multi-part questions
- Handling of ambiguous queries and edge cases

3. Explainability (20%)

- Clarity and completeness of explanations
- Transparency in reasoning and methodology
- Appropriate use of statistics and evidence

4. Conversational Quality (15%)

- Naturalness of responses

- Context retention across multiple queries
- User experience and ease of interaction

5. Innovation & Technical Implementation (10%)

- Novel approaches to insight generation
- Effective use of AI/ML techniques
- System architecture and scalability considerations

7. Rules and Constraints

Mandatory Requirements

1. **Dataset Usage**: Teams **must use only the organizer-provided synthetic dataset**. Use of external transaction datasets is not permitted.
2. **Data Integrity**: The provided dataset must not be modified structurally (columns cannot be removed or renamed). Derived features and transformations are allowed.
3. **Technology Stack**: Teams are free to choose any programming languages, frameworks, libraries, or AI models (including LLMs, traditional ML, or rule-based systems).
4. **Focus Area**: While a basic user interface is expected, the primary evaluation focus is on **insight quality, accuracy, and explainability** rather than UI aesthetics.
5. **Originality**: All code and solutions must be original work developed during the hackathon period.

Prohibited Actions

- Using real-world payment data or datasets other than the one provided
- Hardcoding answers to specific queries
- Plagiarizing code or solutions from external sources
- Collaboration between teams

8. Deliverables

Teams must submit:

1. **Source Code**: Complete, documented codebase with clear instructions for setup and execution
2. **Working Demo**:
   - A functional prototype (web application, command-line interface, or notebook)
   - Video demonstration (3-5 minutes) showing the system handling diverse queries
3. **Technical Documentation**:
   - System architecture overview
   - Approach to query understanding and insight generation
   - Data analysis methodology
   - Key technical decisions and justifications
   - Setup and installation instructions
4. **Sample Query Set**:
   - At least 15 diverse example queries the system can handle
   - Corresponding system responses demonstrating capabilities
5. **Presentation**:
   - 10-minute pitch deck covering problem understanding, approach, demo, and impact

Conclusion

This challenge offers participants an opportunity to work at the intersection of natural language processing, data analytics, and business intelligence. By building a system that makes transaction data accessible through conversation, teams will contribute to democratizing data-driven decision-making in the digital payments industry. The best solutions will combine technical sophistication with practical utility, delivering insights that business leaders can immediately understand and act upon.

Structure of the competition

Round 1: Concept & Approach Submission

Teams must submit a structured concept response through **Google Form**, outlining their understanding of the problem, the leadership-level questions they aim to

address, their high-level analytical approach, assumptions, and explainability strategy. No dataset access or coding is required at this stage. Submissions will be evaluated using a predefined rubric, and shortlisted teams will advance to the final round.

Round 2: Final Demo & Presentation

Shortlisted teams will receive access to the dataset and are required to build a working solution that generates data-backed, explainable insights. Teams will present their approach and demonstrate their system to the judging panel, followed by a Q&A. Evaluation will focus on insight correctness, clarity, depth of analysis, and practical relevance.

Timeline

Registration Deadline	10th Feb 2026
Abstract Submission Deadline	10th Feb 2026
Round 1 Result Announcement	13th Feb 2026
Round 2 Submission Deadline	28th Feb 2026
Final Presentation	8th Mar 2026

Team Structure

- Participation is **team-based only**.
- Each team must consist of **2 to 4 members**.
- Team members may belong to the **same or different institutions**.
- A participant may be part of **only one team**. Multiple team entries by the same participant are not allowed.

Prize Money

Prize money will be awarded to the top three teams per theme via NEFT by the latest May 2026.